Unit 2

TECHNOLOGY AND INCENTIVES

OUTLINE

A. Introduction
B. Decision making
C. Technologies and innovation
D. Economic models
E. Capitalism and climate change

A. Introduction

The Context for This Unit

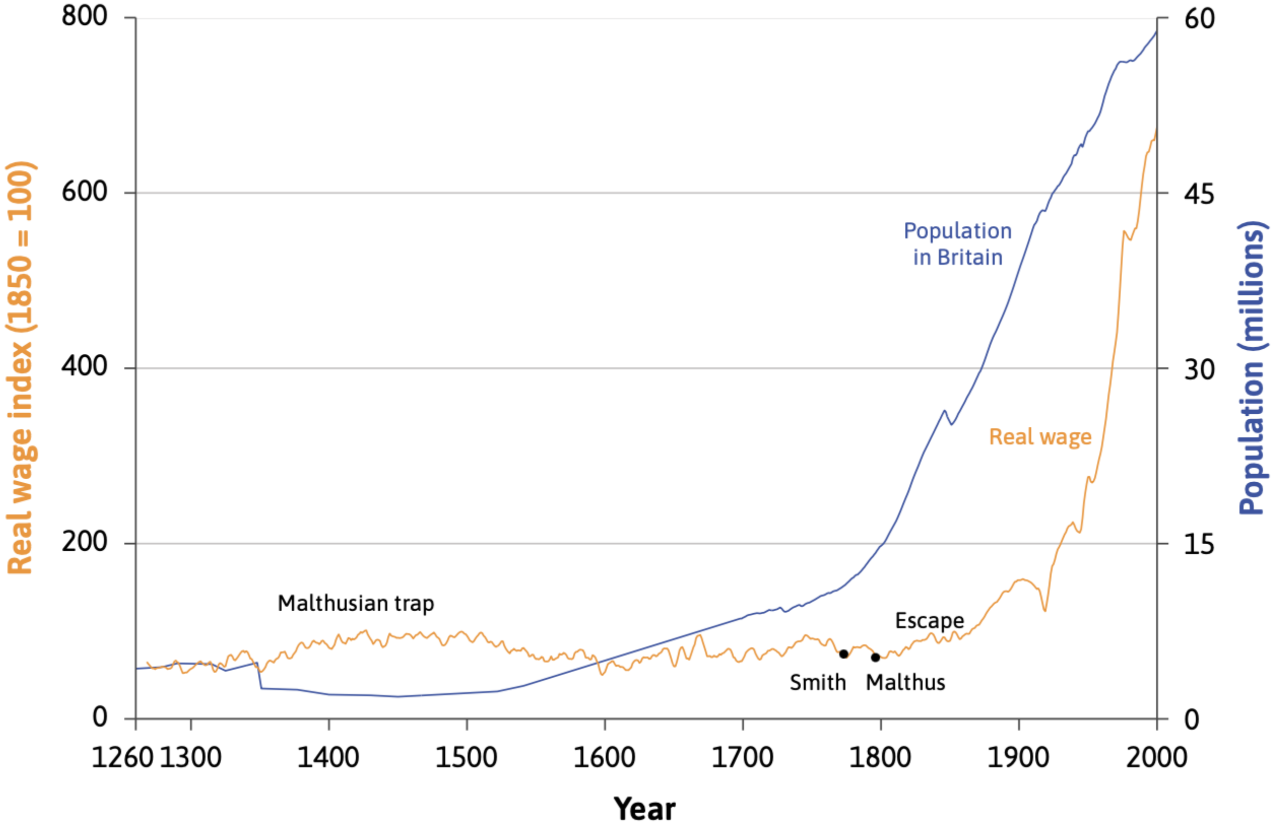
The recent rapid, sustained increase in income and living standards is largely due to technological progress.

(Unit 1)

However, these major changes started very suddenly, 200 years ago.

- How did the technological revolution start?
- Why did it not start earlier?

This Unit



Use economic models to explain the rapid growth in real wages and population in the last 2 centuries.

B. Decision making

A model of decision making

- **Opportunity cost** – the value of the next best action not taken
- **Reservation option** – the next best alternative
- **Economic cost** = direct costs incurred by taking an action + opportunity cost
- **Economic rent** = net benefit from option taken – opportunity cost

Scenario: Are you going to the concert?

- A ticket costs \$25
- Enjoyment value of concert is \$55
- Net benefit of concert = \$55 - \$25 = \$30

- Reservation option is earning \$22 for babysitting
- Opportunity cost = \$22

- Economic cost of concert = \$25 + \$22 = \$47
- Economic rent = \$55 - \$47 = \$8

Innovation rent and relative prices

- **Innovation rents** – a form of economic rent. The extra profits made by exploiting an invention. Provide **incentives** for taking action.

- **Relative prices** – the price of one option relative to another, often expressed as a ratio of the two prices. An important factor in determining economic **incentives**.

C. Technologies and innovation

Specialization

Specialization means each person does just one or a few of the large number of tasks necessary for some project or goal.

Specialization can happen for three reasons:

- Learning by doing
- Difference in ability
- **Economies of scale**

Comparative and absolute advantage

	Production if 100% of time is spent on one good
Greta	1,250 apples or 50 tonnes of wheat
Carlos	1,000 apples or 20 tonnes of wheat

- Being more productive in producing a good means having an **absolute advantage.**

- Having a lower relative cost of producing a good means having a **comparative advantage.**

Comparative and absolute advantage

	Opportunity cost of 1 tonne of wheat	Opportunity cost of 1 apple
Greta	25 apples	0.04 tonnes of wheat
Carlos	50 apples	0.02 tonnes of wheat

- Being more productive in producing a good means having an **absolute advantage.**
- Having a lower relative cost of producing a good means having a **comparative advantage.**

Self-sufficiency vs specialization and trade

In self-sufficiency:

- Greta uses 40% of her time on apple production and 60% on wheat
- Carlos spends 30% on apple production and 70% on wheat

		Self-sufficiency	Complete specialization and trade		
			Production	Consumption	Trade
		(1)	(2)	(3)	(4)
Greta	Apples	500	0	600	Buys 600 apples
	Wheat	30	50	35	Sells 15t wheat
Carlos	Apples	300	1,000	400	Sells 600 apples
	Wheat	14	0	15	Buys 15t wheat
Total	Apples	800	1,000	1,000	
	Wheat	44	50	50	

Firms, technology and production

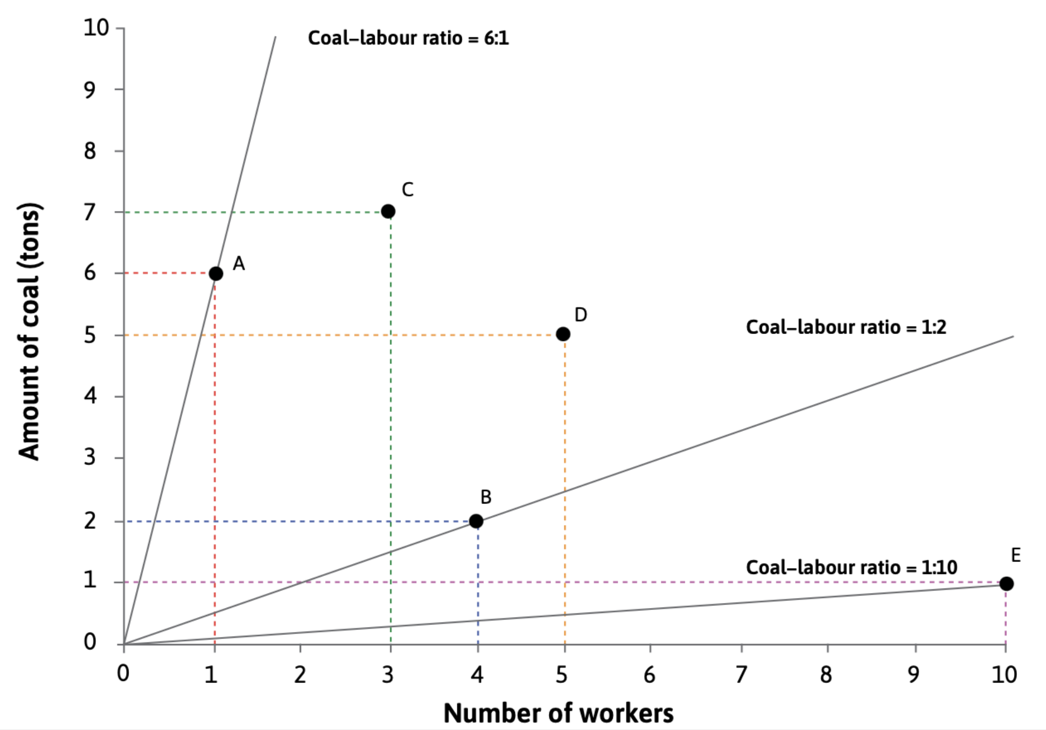
- **Division of labour** – specialisation in producing things either within a firm or across society
- **Production technology** - the process that a firm uses to convert a set of inputs into an output that it can sell.
- **Factors of production** – the inputs into the production process (e.g. raw materials, labour, capital goods, energy)
- **Production function** – a relationship that tell us how much output will be produced given the amounts of inputs used

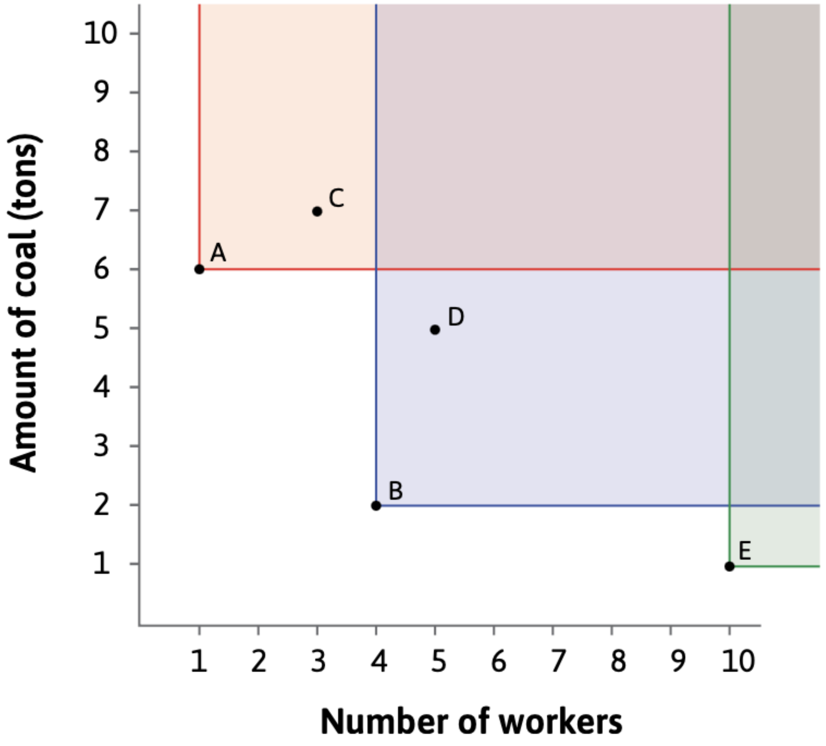
Fixed-proportions technology and CRS

Number of machines, *M*	Number of workers, *N*	Amount of energy, *E* (kWh)	Output, *Y* (litres)
3	1	80	50
6	2	160	100
9	3	240	150
12	4	320	200

Which technology?

Suppose there are 5 possible technologies for producing 100 metres of cloth. Which one should a firm choose?



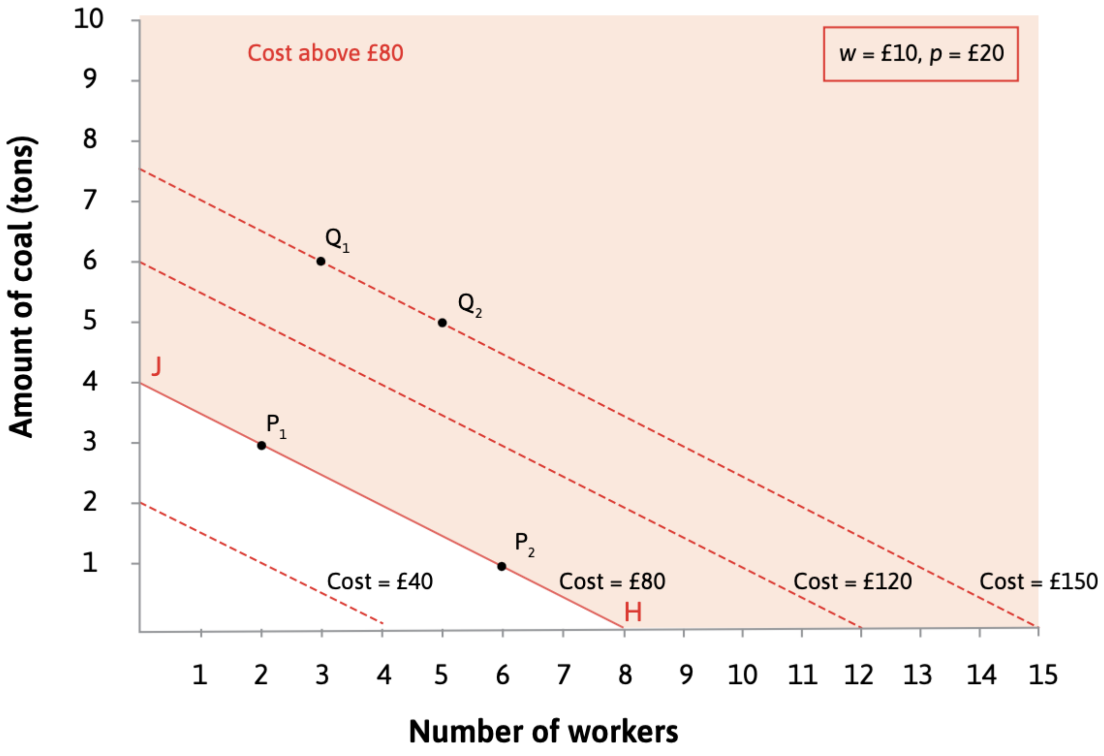


Which technology?

- Assume the motive of the firm is to maximise profit. This means they will want to minimise the cost. Also, assume w = £10 and p = £20.

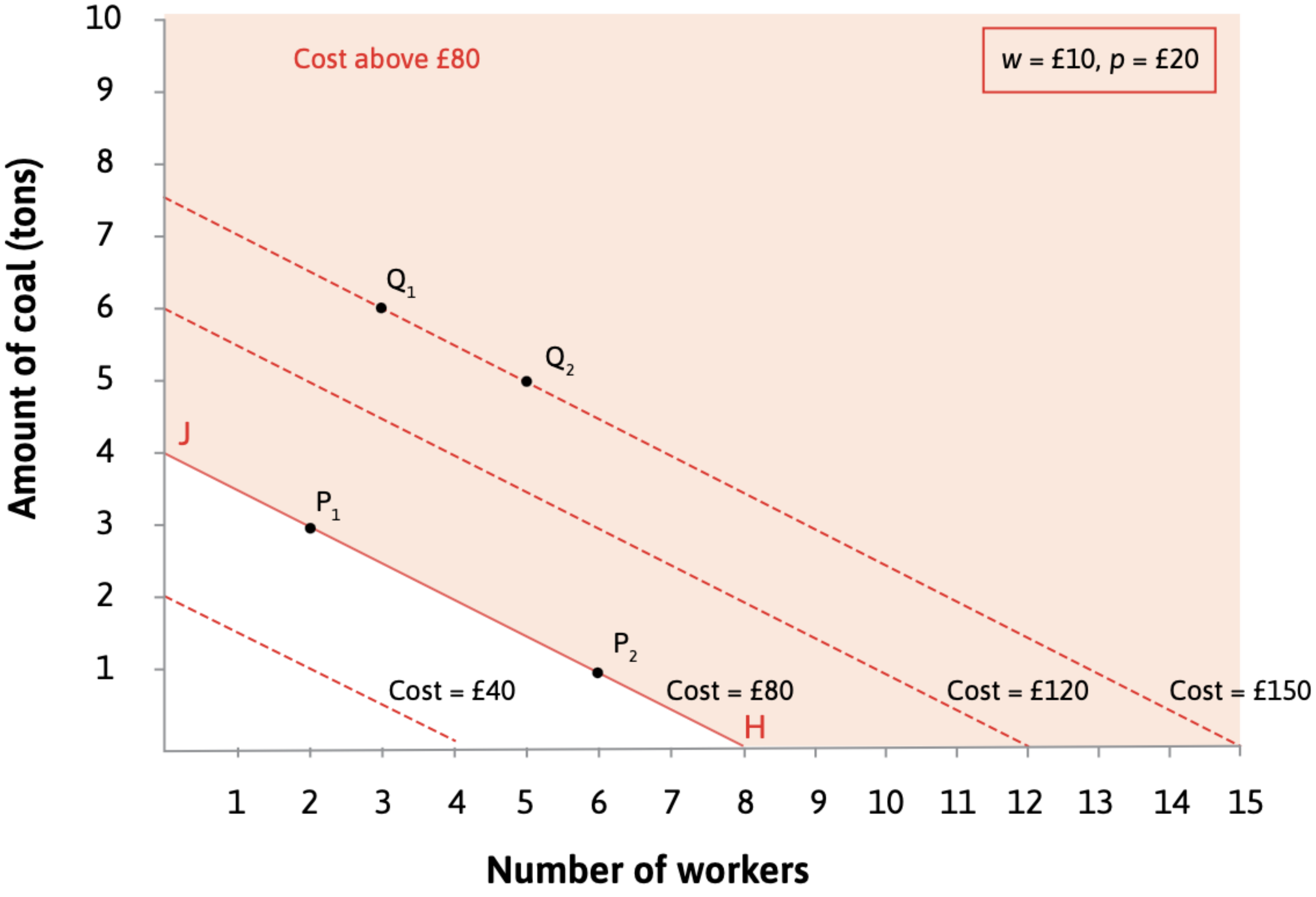
Cost = (wage × workers) + (price of a ton of coal × number of tons)

$$= (w \times N) + (p \times R)$$

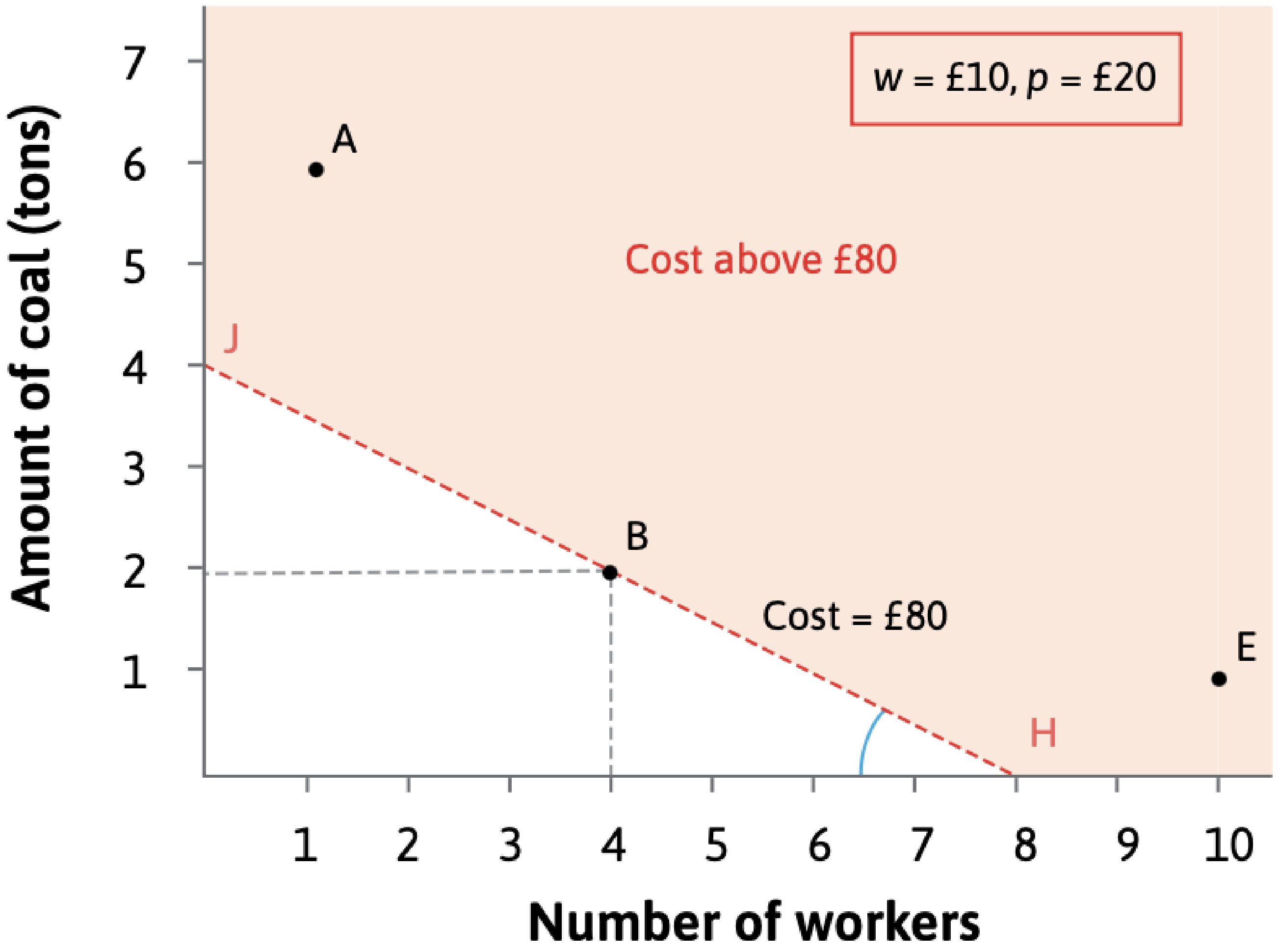


Which technology?

- Isocost lines join all the combinations of workers and coal that cost the same amount
- Slope of isocost lines = -w/p
- Slope depends on <u>relative price</u>



Which technology?



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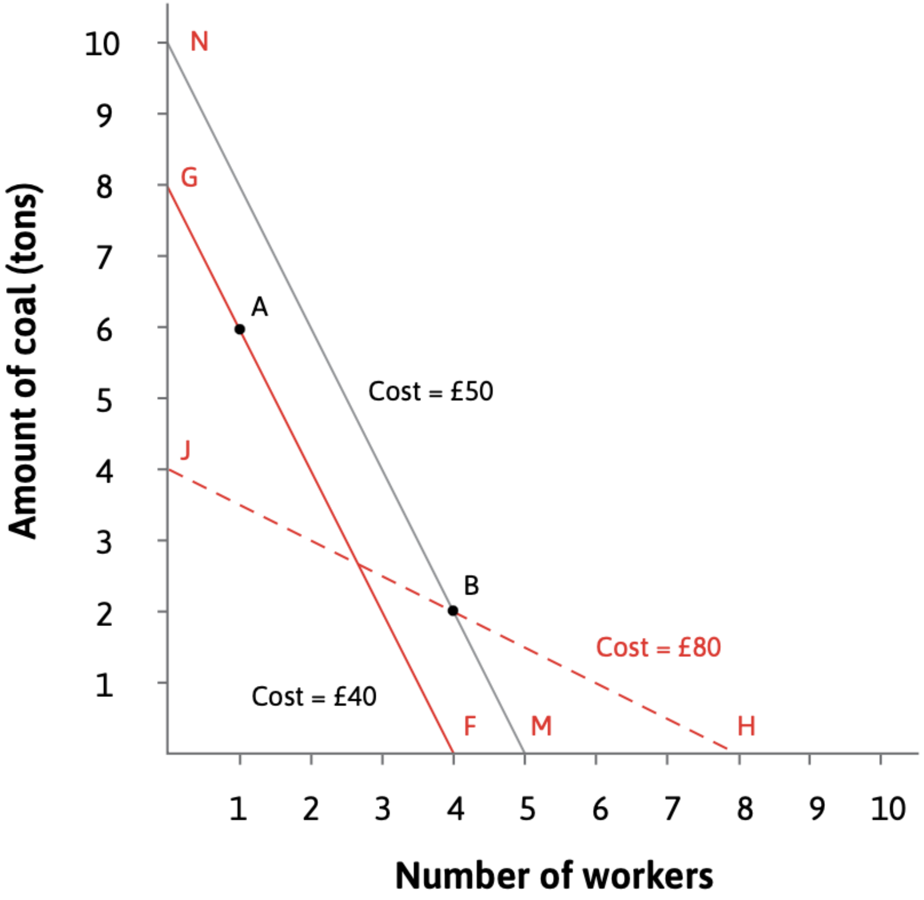
A change in relative prices

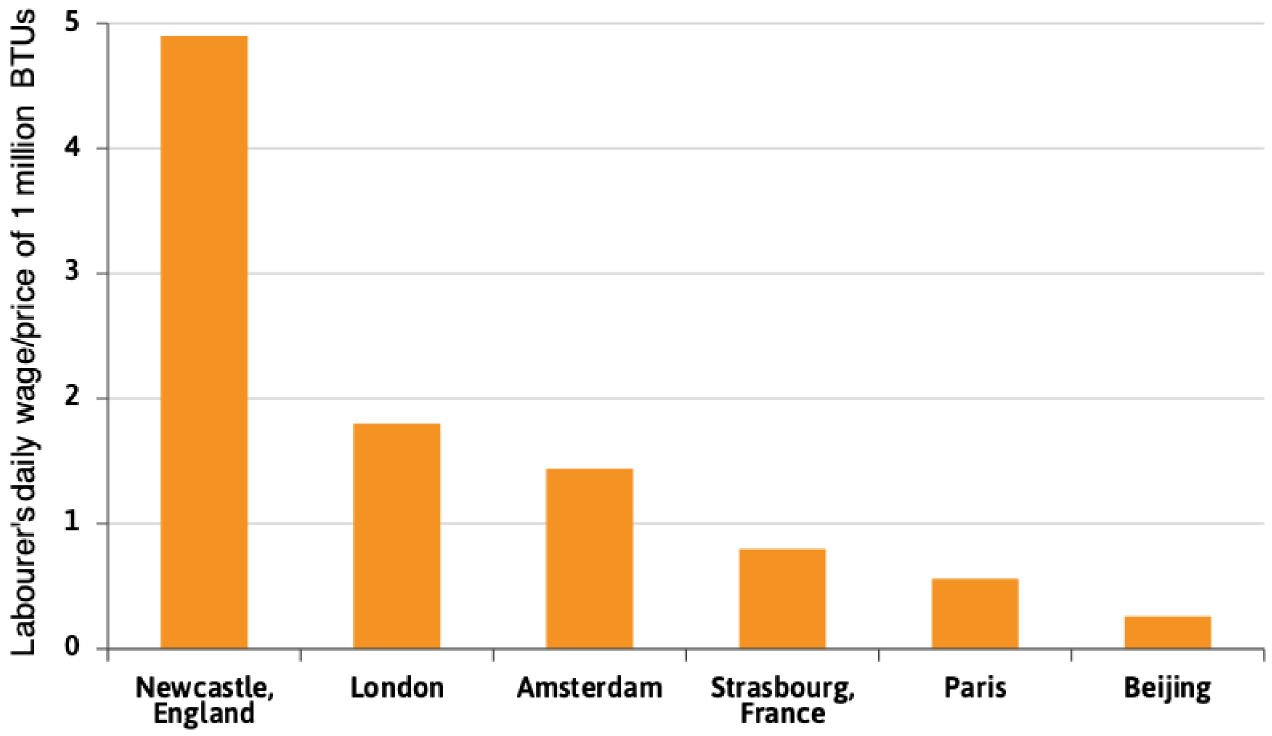
- Suppose the price of coal falls to £5, the wage remains at £10 and the price of cloth stays the same.



The Industrial Revolution in Britain

The price of labour relative to energy was very high in Britain for two reasons:

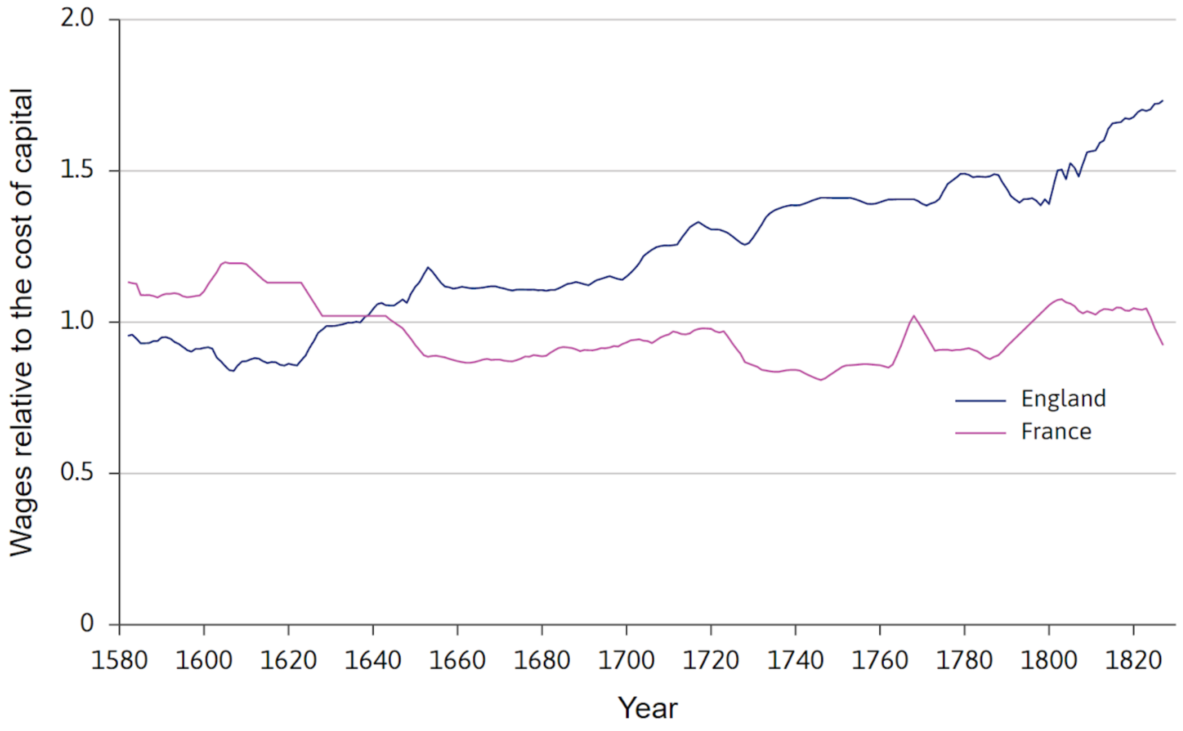
- English wages were higher than wages elsewhere
- Coal was cheaper in coal-rich Britain than in the other countries



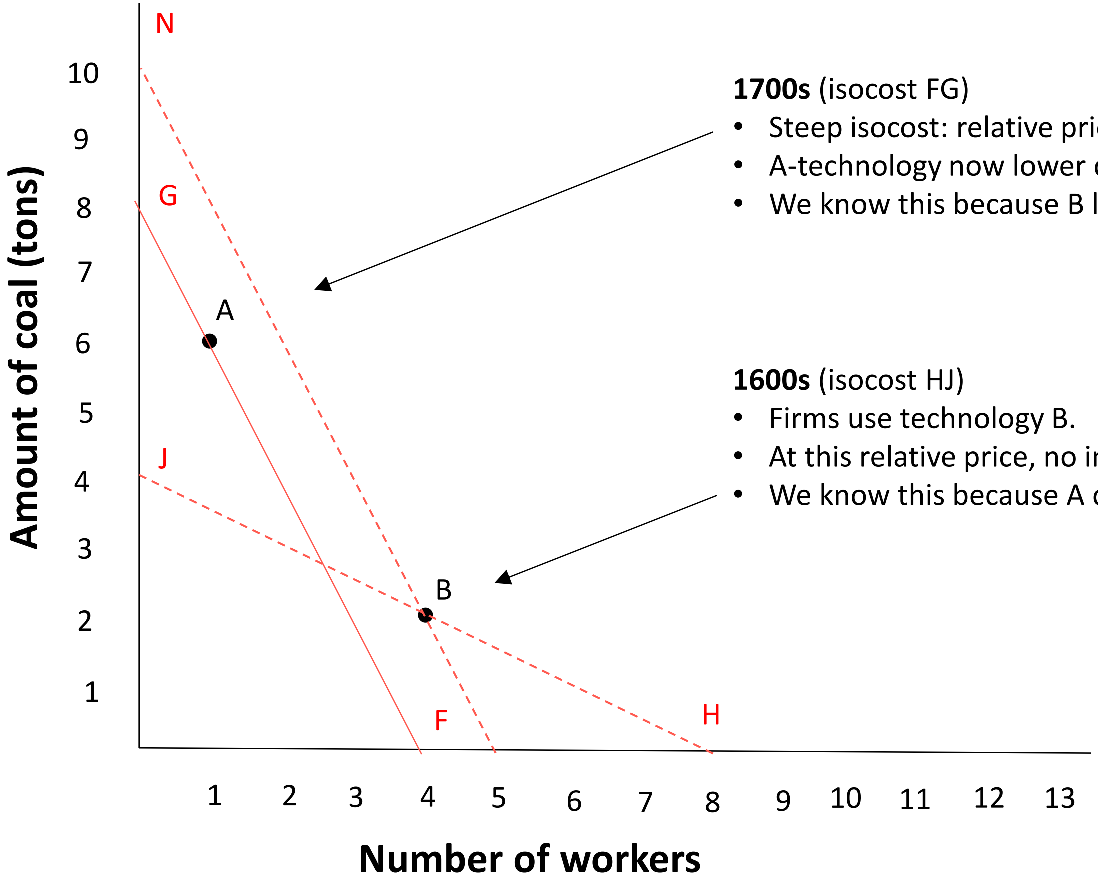
The Industrial Revolution in Britain

Wages relative to the costs of both energy and capital goods

- rose in the eighteenth century in Britain compared with earlier historical periods;
- and were higher in Britain during the eighteenth century than elsewhere.



The Industrial Revolution in Britain



1700s (isocost FG)

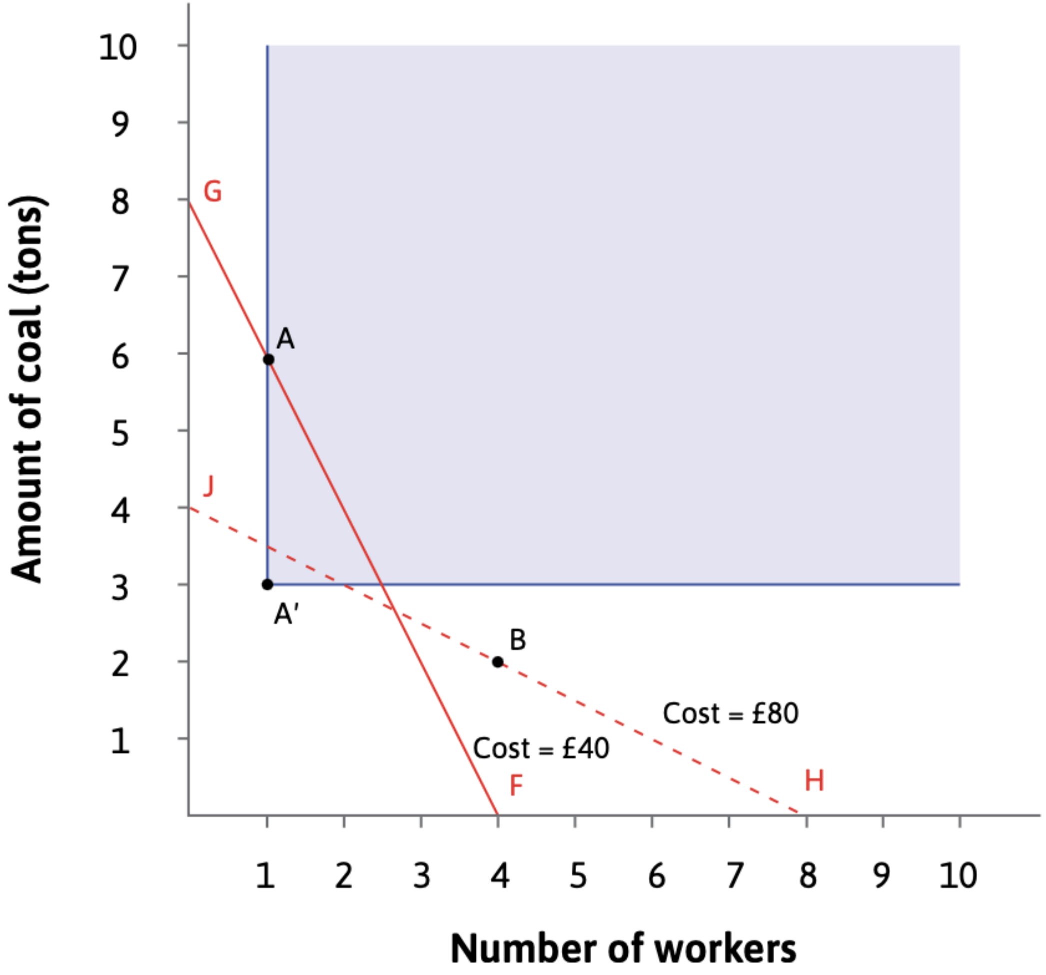
- Steep isocost: relative price of labour to coal is high.
- A-technology now lower cost than the B-technology
- We know this because B lies outside the line FG.

1600s (isocost HJ)

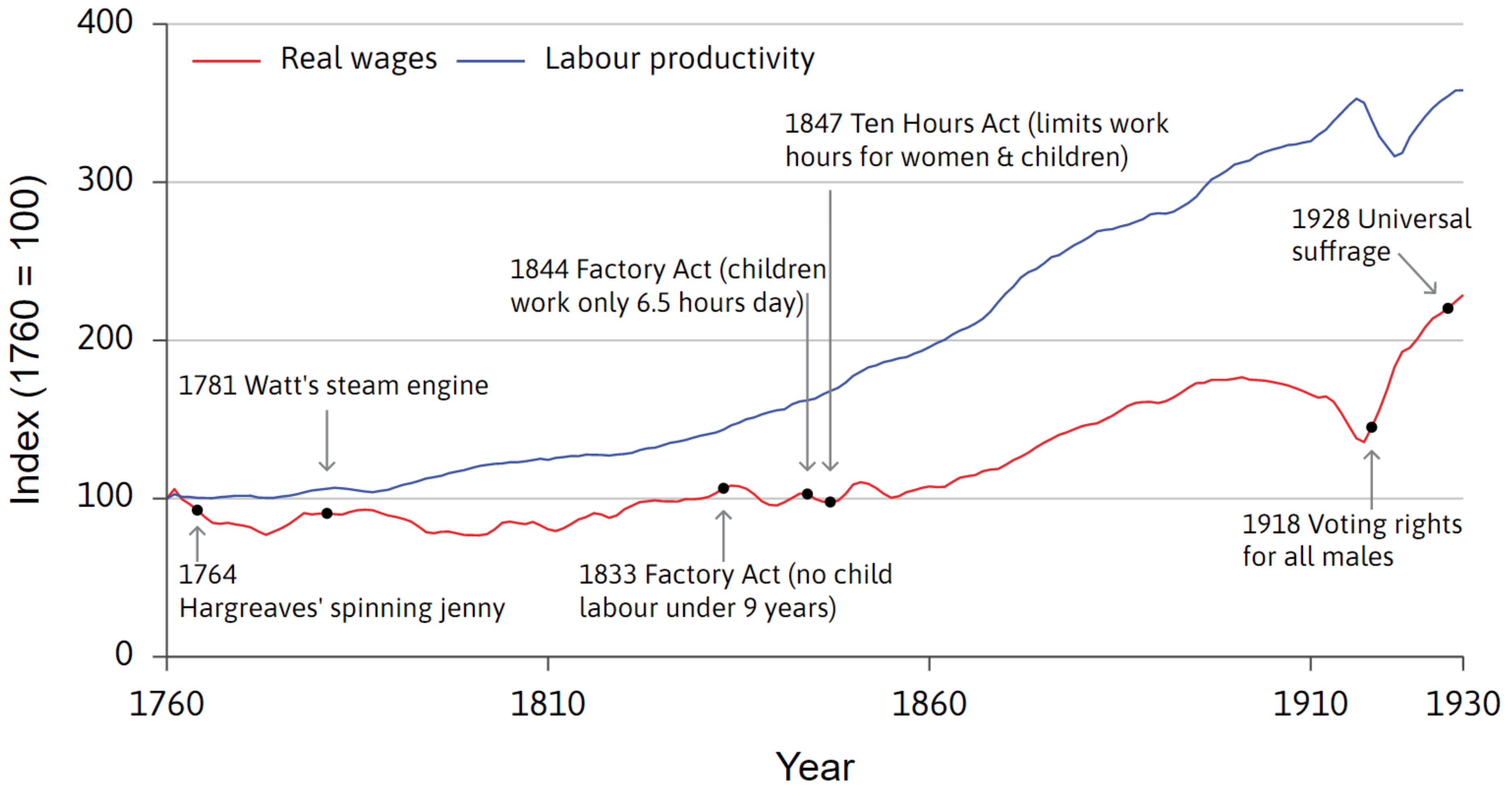
- Firms use technology B.
- At this relative price, no incentive to develop technology A.
- We know this because A costs more (it lies outside the line HJ)

The spread of technology

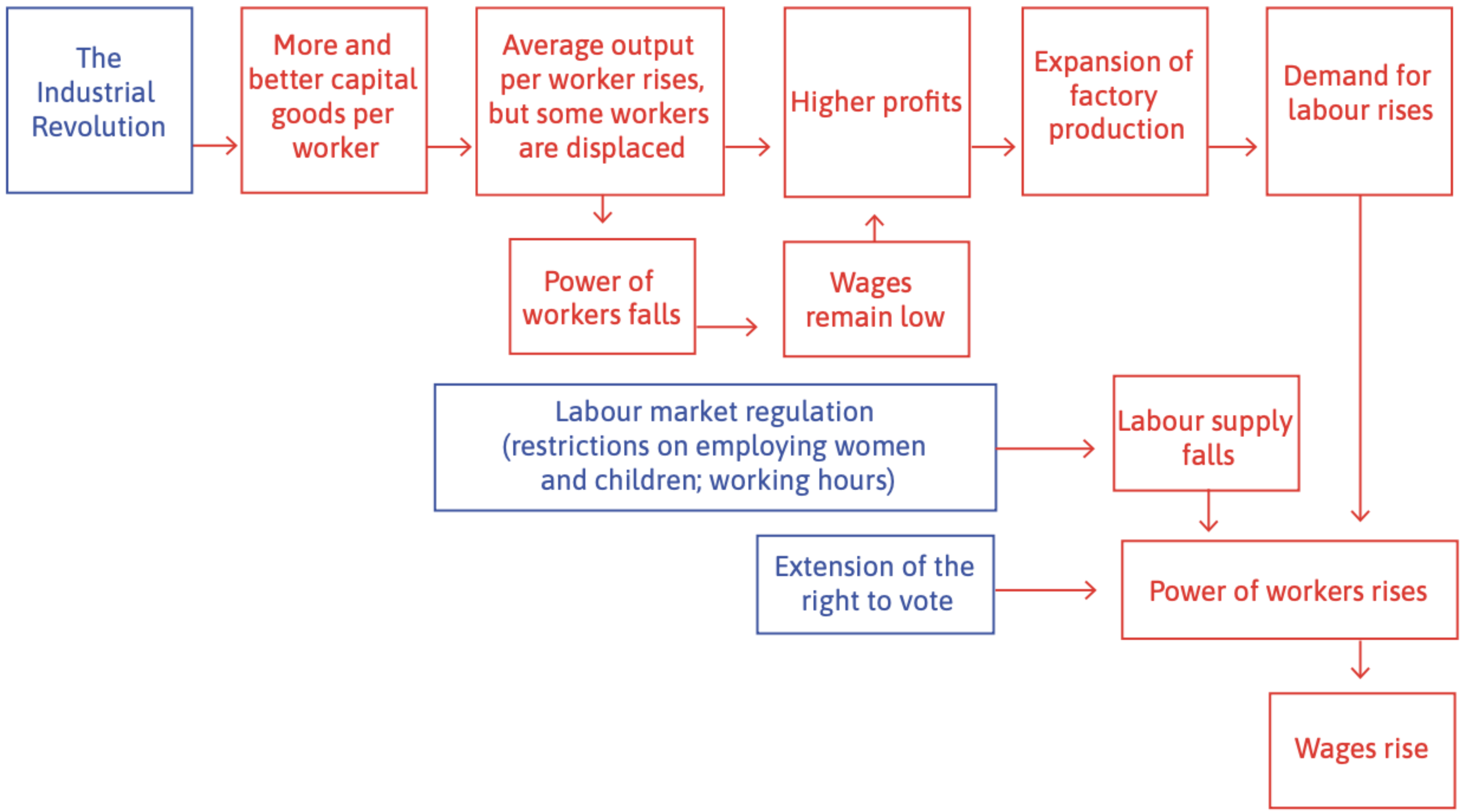
- Further technological progress
- Wage growth
- Falling energy costs



Escaping the Malthusian trap



Escaping the Malthusian trap



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D. Economic models

Model-building

When we build a model, the process follows these steps:

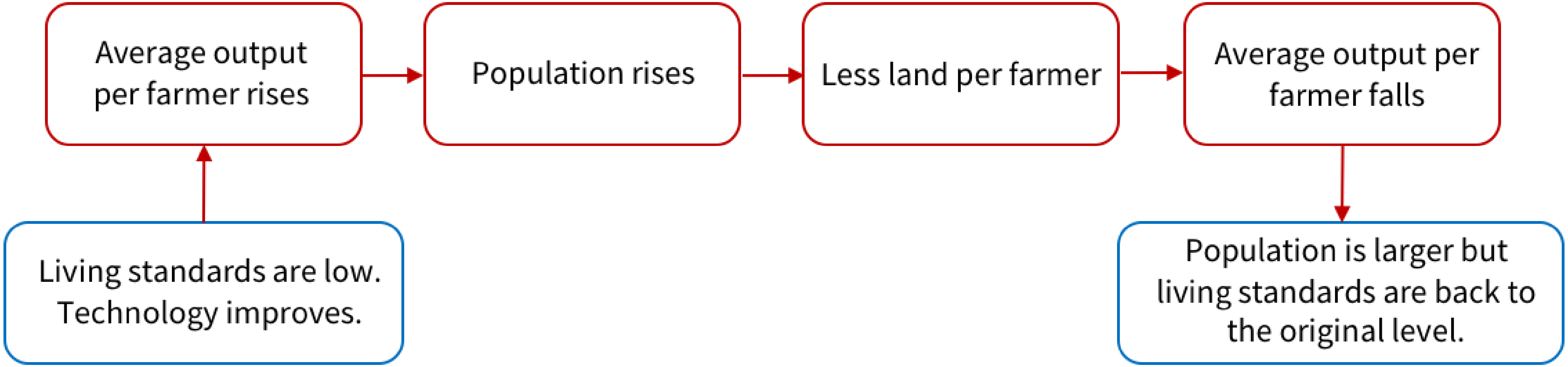
1. We begin with a well-defined question.
2. We construct a simplified description of the conditions under which people take actions.
3. We describe in simple terms what determines the actions that people take.
4. We determine how each of their actions affects each other.
5. We determine the outcome of these actions. This is often an equilibrium (something is constant).
6. Finally, we try to get more insight by studying what happens to certain variables when conditions change.

Key ideas in a model

- **Equilibrium** – a situation that is self-perpetuating, meaning that there is no tendency for the situation to change unless an external force for change is introduced
- **Endogenous variables** – variables whose values are determined by relationships built into the mode
- **Exogenous variables** – variables whose values are determined outside the model
- **Ceteris paribus** – holding other things equal

Example: The Malthusian model

Why technological improvement in farming doesn't raise living standards

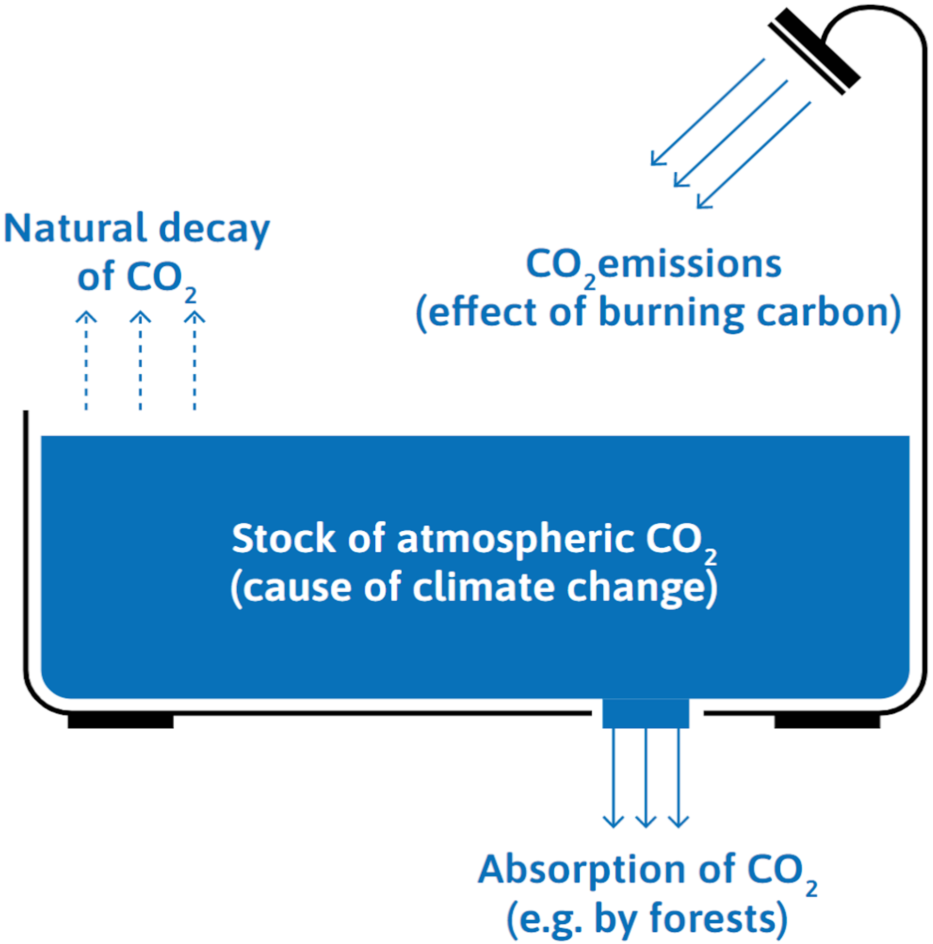


E. Capitalism and climate change

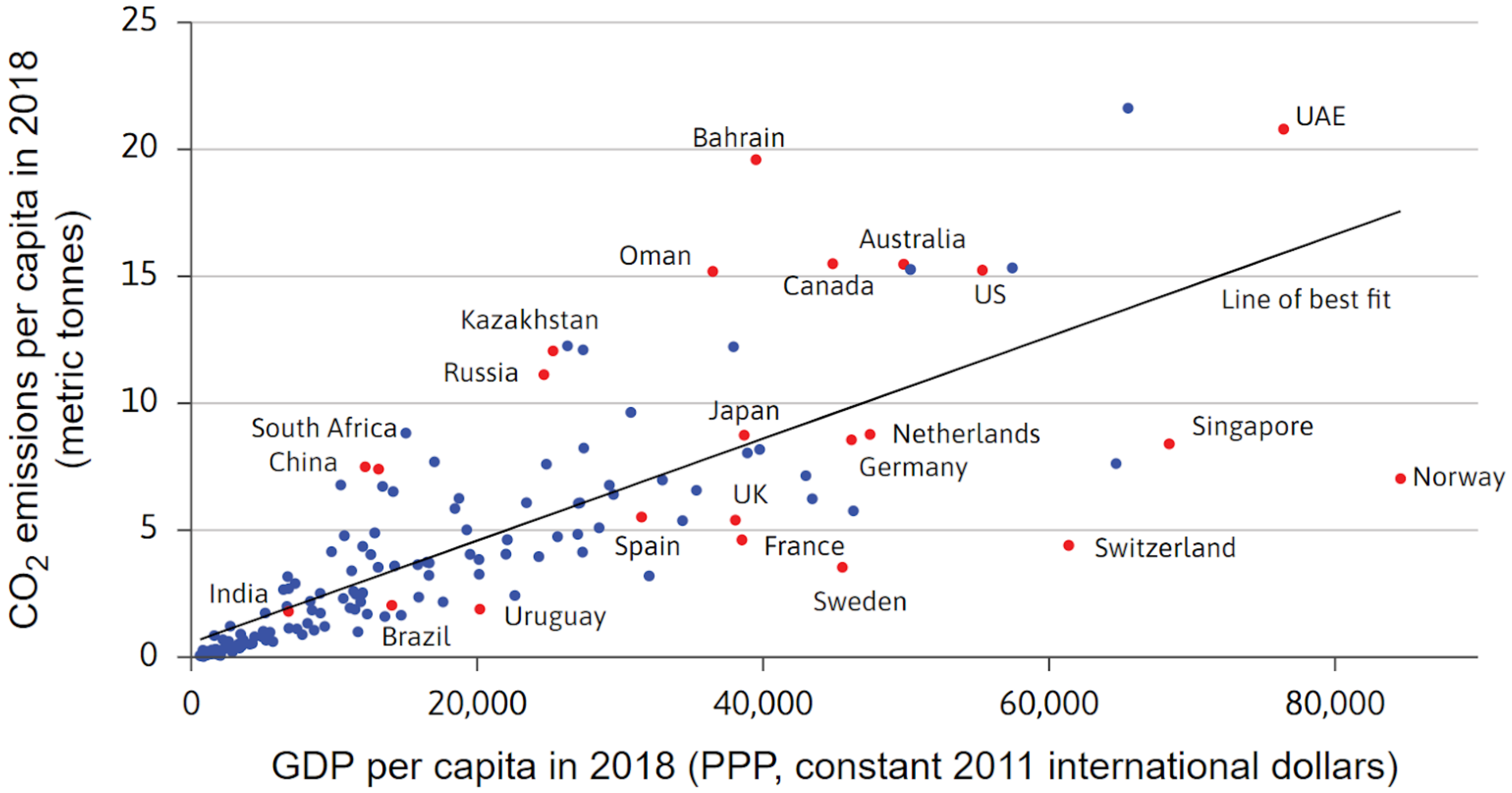
Capitalism + carbon = hockey-stick growth + climate change

- The Industrial Revolution marked the transition from an economy in which photosynthesis was the source of most energy, so that land is a constraint on growth, to an energy-rich economy based on fossil fuels.
- It led to unprecedented increases in per capita income.
- It also led to unprecedented increases in the surface temperature of the Earth.
- The ongoing reduction in global poverty cannot be accomplished by the same coal-plus-capitalism

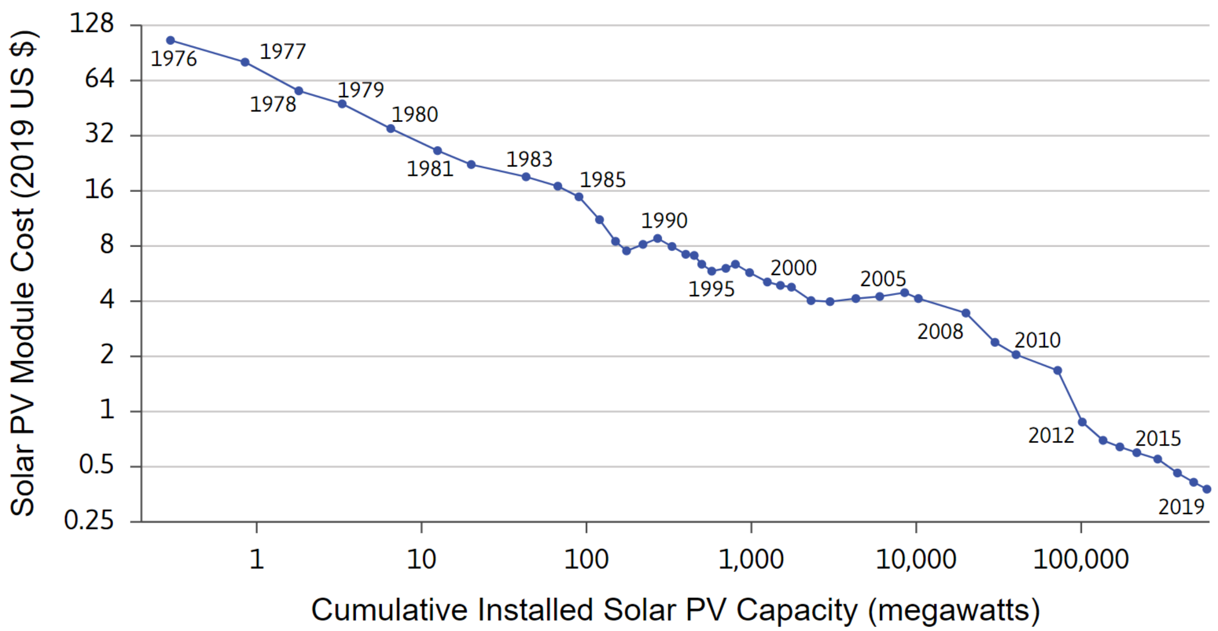
The basic science of climate change



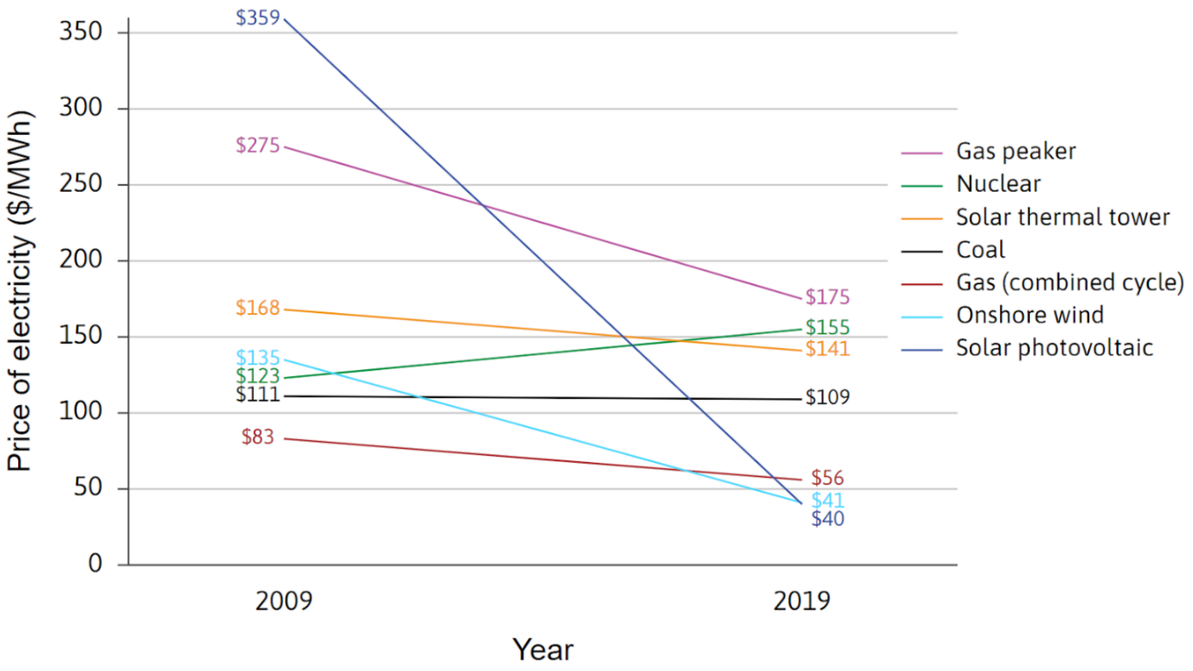
An unsustainable path



Some hope?

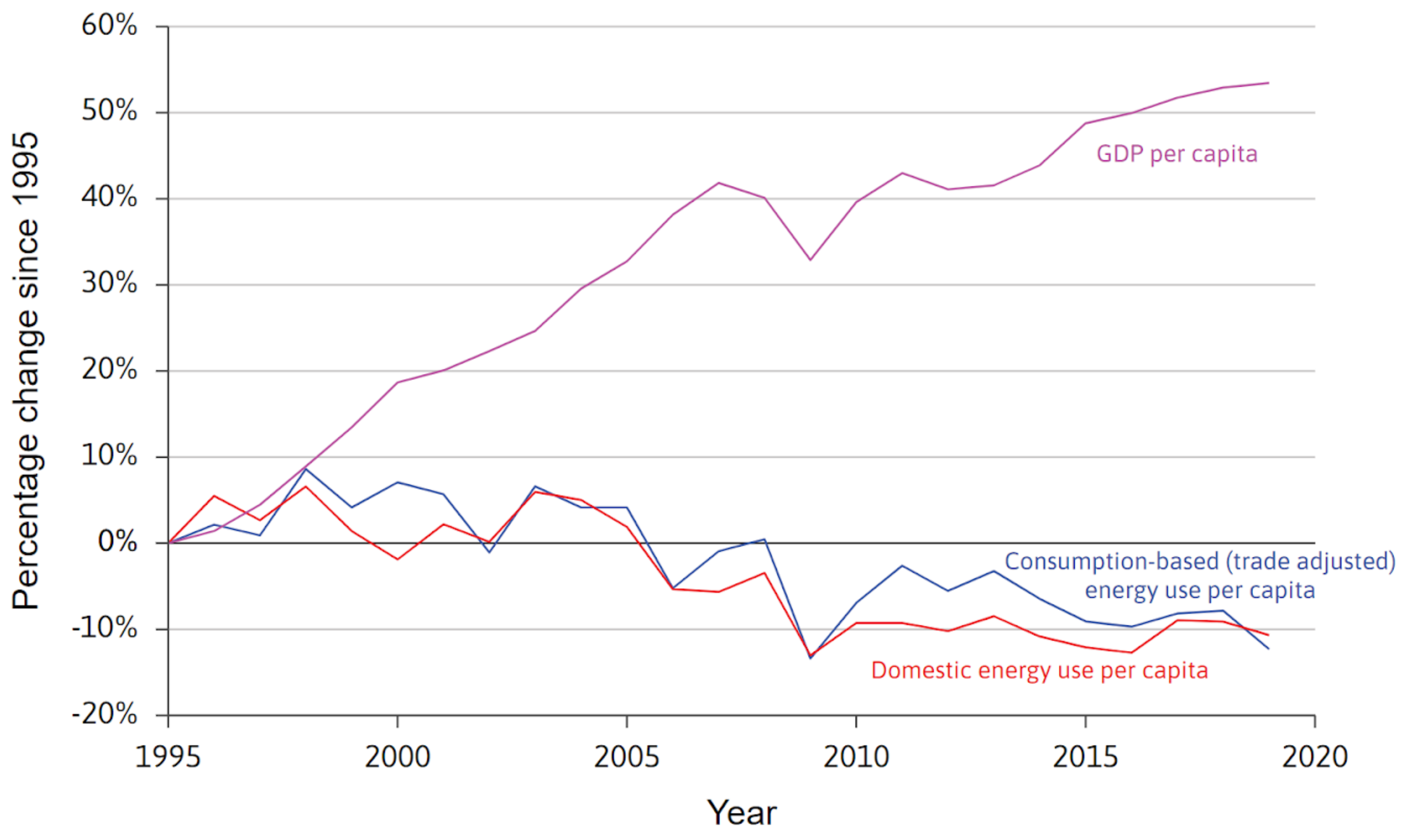


Solar PV Module Cost (2019 US $)
128
64
32
16
8
4
2
1
0.5
0.25
1976
1977
1978
1979
1980
1981
1983
1985
1990
1995
2000
2005
2008
2010
2012
2015
2019
1
10
100
1,000
10,000
100,000
Cumulative Installed Solar PV Capacity (megawatts)



Price of electricity ($/MWh)
350
300
250
200
150
100
50
0
$359
$275
$168
$135
$123
$111
$83
$175
$155
$141
$109
$56
$41
$40
2009
2019
Year
Gas peaker
Nuclear
Solar thermal tower
Coal
Gas (combined cycle)
Onshore wind
Solar photovoltaic

Some hope?



60%
50%
40%
30%
20%
10%
0%
-10%
-20%
Percentage change since 1995
GDP per capita
Consumption-based (trade adjusted) energy use per capita
Domestic energy use per capita
1995
2000
2005
2010
2015
2020
Year

Summary

1. Used models for insights on the technological revolution
- Model of a firm: high wages (relative to capital, including energy) motivated **technological innovation**

2. Introduction to economic models
- Less is more: use simplifying assumptions e.g. ceteris paribus

In the next unit

- More about models: An economic model of decision making under constraints
- How individuals respond to technological change: Explaining trends in choices of working hours across time